

Proof of Corollary G.3. Apply Theorem G.2 separately to $c = 0$ and $c = 1$, each with failure probability $\delta/2$. By a union bound, with probability at least $1 - \delta$, the tracking bounds hold simultaneously for both classes and for all $0 \leq t \leq T$.

By definition, we have

$$\begin{aligned}
|\widehat{\mathcal{S}}_t(x) - \mathcal{S}_t(x)| &= |\kappa(\widehat{\boldsymbol{\mu}}_{1,t} - \widehat{\boldsymbol{\mu}}_{0,t})^\top f_{\theta,\mathbf{v}}(x) - \kappa(\boldsymbol{\mu}_{1,t} - \boldsymbol{\mu}_{0,t})^\top f_{\theta,\mathbf{v}}(x)| \\
&\stackrel{(a)}{\leq} \kappa (\|\widehat{\boldsymbol{\mu}}_{0,t} - \boldsymbol{\mu}_{0,t}\| + \|\widehat{\boldsymbol{\mu}}_{1,t} - \boldsymbol{\mu}_{1,t}\|) \\
&\stackrel{(b)}{\leq} 2\kappa \sqrt{(1 - c\rho)^t \cdot \frac{1}{16} + C \left(\sqrt{\rho \log \frac{4T}{\delta}} + \frac{\eta^2}{\rho^2} \right)} \\
&\stackrel{(c)}{\leq} \kappa C \sqrt{(1 - c\rho)^t + \sqrt{\rho \log \frac{4T}{\delta}} + \frac{\eta^2}{\rho^2}}.
\end{aligned}$$

where (a) follows from the triangle inequality and the fact that $\|f_{\theta,\mathbf{v}}(x)\| \leq 1$, (b) applies Theorem G.2, and (c) enlarges the constant C properly. $\square$

Assumption 3 (Local Mass Condition Around the Threshold). *There exist constants $c_0, C_0 > 0$, exponents $\underline{\gamma}, \bar{\gamma} > 0$ satisfying $\underline{\gamma} \leq 1 + \bar{\gamma}$, and $\varepsilon_0 > 0$ such that for all $\varepsilon \in [0, \varepsilon_0]$,*

$$\mathbb{P}_0(|\mathcal{S}(X) - \tau_\alpha^*| \leq \varepsilon) \leq C_0 \varepsilon^{\bar{\gamma}},$$

and

$$\min \{\mathbb{P}_0(\tau_\alpha^* - \varepsilon \leq \mathcal{S}(X) < \tau_\alpha^*), \mathbb{P}_0(\tau_\alpha^* < \mathcal{S}(X) \leq \tau_\alpha^* + \varepsilon)\} \geq c_0 \varepsilon^{\underline{\gamma}}.$$

This assumption imposes regularity conditions on the null score distribution in a neighborhood of τ_α^* . The upper bound controls the concentration of probability mass near the threshold, corresponding to a standard margin-type condition in classification theory [81–83, 28]. The lower bound ensures that there is sufficient probability mass around the threshold, preventing the distribution from being too sparse in this region.

Theorem G.4 (Formal Statement of Theorem 3.3). *Let Assumptions 1-3 hold, and suppose that the conditions of Theorem G.2 hold for both classes with failure probability $\delta/2$. Suppose further that the locality condition in Lemma G.6 holds. Then there exists a constant $c \in (0, 1)$ such that, with probability at least $1 - 2\delta$ over the randomness in the training procedure and the calibration sample, the excess Type-II error satisfies*

$$\mathbb{P}_1(\widehat{\mathcal{R}}_t) - \mathbb{P}_1(\mathcal{R}_t^*) \leq \underbrace{\mathcal{O} \left(\left((1 - c\rho)^t + \sqrt{\rho \log(4T/\delta)} + \eta^2/\rho^2 \right)^{\frac{1+\bar{\gamma}}{2}} \right)}_{\text{Estimation Error}} + \underbrace{\sqrt{\frac{\log(2/\delta)}{n_2}} + \frac{1}{n_2}}_{\text{Calibration Error}}$$

where $t \in [1, T]$ is the iteration index within the total horizon T , $\rho \in (0, 1)$ is the EMA coefficient satisfying $\rho \leq 1/c$, $\eta > 0$ is the steering learning rate, n_2 is the calibration sample size, $\delta \in (0, 1)$ is the confidence parameter, and $\underline{\gamma}, \bar{\gamma}$ are the constants in Assumption 3.

G.3 Proof of Theorem G.4

Proof of Theorem G.4. For notational simplicity, we suppress the time index t . The excess Type-II error admits the decomposition given in Lemma G.5. We prove Lemma G.5 in Appendix G.5.

Lemma G.5. *Let p_0 and p_1 denote the density functions of $\mathbb{P}_0$ and $\mathbb{P}_1$, respectively. Define the score function $\mathcal{S}(x) = \log \frac{p_1(x)}{p_0(x)}$, and the acceptance regions*

$$\mathcal{R}^* = \{x : \mathcal{S}(x) \leq \tau_\alpha^*\}, \quad \widehat{\mathcal{R}} = \{x : \widehat{\mathcal{S}}(x) \leq \widehat{\tau}_\alpha\}.$$

Then,

$$\mathbb{P}_1(\widehat{\mathcal{R}}) - \mathbb{P}_1(\mathcal{R}^*) = \underbrace{\int_{\widehat{\mathcal{R}} \Delta \mathcal{R}^*} \left| \frac{p_1(x)}{p_0(x)} - e^{\tau_\alpha^*} \right| d\mathbb{P}_0(x)}_{\text{Estimation Error}} + \underbrace{e^{\tau_\alpha^*} \left(\alpha - \mathbb{P}_0(\widehat{\mathcal{R}}^c) \right)}_{\text{Price of Conservativeness}}. \quad (9)$$

Here, $\widehat{\mathcal{R}} \Delta \mathcal{R}^*$ denotes the symmetric difference, i.e., $\widehat{\mathcal{R}} \Delta \mathcal{R}^* = (\widehat{\mathcal{R}} \setminus \mathcal{R}^*) \cup (\mathcal{R}^* \setminus \widehat{\mathcal{R}})$.

From (9), there are two terms in the decomposition of the excess Type-II error.

(i) The second term quantifies the gap between empirical Type I error and the target level α .

By Theorem 3.1, with probability at least $1 - \delta$, we have $|\mathbb{P}_0(\widehat{\mathcal{R}}^c) - \alpha| \leq \Delta_{n_2}(\delta)$ with $\Delta_{n_2}(\delta) := \sqrt{\frac{\log(2/\delta)}{2n_2}} + \frac{1}{n_2}$, as a result of which,

$$e^{\tau_\alpha^*} (\alpha - \mathbb{P}_0(\widehat{\mathcal{R}}^c)) \leq e^{\tau_\alpha^*} \Delta_{n_2}(\delta). \quad (10)$$

(ii) The first term on the RHS of (9) is due to the estimation error between the oracle score $\mathcal{S}$ and the estimated $\widehat{\mathcal{S}}$. To bound it, we have to bound the discrepancy of the acceptance regions in the score space. By Corollary G.3, with probability at least $1 - \delta$, we have $\sup_{x \in \mathcal{X}} |\widehat{\mathcal{S}}(x) - \mathcal{S}(x)| \leq r(\delta, \rho, \eta)$.

Since both μ_c and $f_{\theta, \mathbf{v}}(x)$ lie on the unit hypersphere, the log-scores are deterministically bounded within $[-2\kappa, 2\kappa]$. Therefore, the exponential mapping is Lipschitz continuous with constant $e^{2\kappa}$ over this domain. For any $x \in \widehat{\mathcal{R}}\Delta\mathcal{R}^*$, it follows that

$$\left| \frac{p_1(x)}{p_0(x)} - e^{\tau_\alpha^*} \right| = |e^{\mathcal{S}(x)} - e^{\tau_\alpha^*}| \leq e^{2\kappa} |\mathcal{S}(x) - \tau_\alpha^*|. \quad (11)$$

Lemma G.6. *Suppose*

$$|\mathbb{P}_0(\widehat{\mathcal{R}}^c) - \alpha| \leq \Delta_{n_2}(\delta), \quad \sup_{x \in \mathcal{X}} |\widehat{\mathcal{S}}(x) - \mathcal{S}(x)| \leq r(\delta, \rho, \eta),$$

and

$$2r(\delta, \rho, \eta) + \left(\frac{\Delta_{n_2}(\delta)}{c_0} \right)^{1/\gamma} \leq \varepsilon_0.$$

Then, under Assumption 3,

$$\widehat{\mathcal{R}}\Delta\mathcal{R}^* \subseteq \left\{ x : |\mathcal{S}(x) - \tau_\alpha^*| < 2r(\delta, \rho, \eta) + \left(\frac{\Delta_{n_2}(\delta)}{c_0} \right)^{1/\gamma} \right\}. \quad (12)$$

Consequently,

$$\mathbb{P}_0(\widehat{\mathcal{R}}\Delta\mathcal{R}^*) \leq C_0 \left[2r(\delta, \rho, \eta) + \left(\frac{\Delta_{n_2}(\delta)}{c_0} \right)^{1/\gamma} \right]^{\bar{\gamma}}. \quad (13)$$

Now, we are ready to analyze the first term.

$$\begin{aligned} & \int_{\widehat{\mathcal{R}}\Delta\mathcal{R}^*} \left| \frac{p_1(x)}{p_0(x)} - e^{\tau_\alpha^*} \right| d\mathbb{P}_0(x) \\ & \stackrel{(11)}{\leq} e^{2\kappa} \int_{\widehat{\mathcal{R}}\Delta\mathcal{R}^*} |\mathcal{S}(x) - \tau_\alpha^*| d\mathbb{P}_0(x) \\ & \stackrel{(12)}{\leq} e^{2\kappa} \left[2r(\delta, \rho, \eta) + \left(\frac{\Delta_{n_2}(\delta)}{c_0} \right)^{1/\gamma} \right] \mathbb{P}_0(\widehat{\mathcal{R}}\Delta\mathcal{R}^*) \\ & \stackrel{(13)}{\leq} e^{2\kappa} C_0 \left[2r(\delta, \rho, \eta) + \left(\frac{\Delta_{n_2}(\delta)}{c_0} \right)^{1/\gamma} \right]^{1+\bar{\gamma}}. \end{aligned} \quad (14)$$

With probability at least $1 - 2\delta$, we have both Theorem 3.1 and Corollary G.3 hold. Conditioned on this joint event, combining (10) and (14) with the decomposition (9) yields

$$\begin{aligned} & \mathbb{P}_1(\widehat{\mathcal{R}}_t) - \mathbb{P}_1(\mathcal{R}^*) \\ & \leq e^{2\kappa} C_0 \left[2r_t(\delta, \rho, \eta) + \left(\frac{1}{c_0} \left(\sqrt{\frac{\log(2/\delta)}{2n_2}} + \frac{1}{n_2} \right) \right)^{1/\gamma} \right]^{1+\bar{\gamma}} + e^{\tau_{\alpha,t}^*} \left(\sqrt{\frac{\log(2/\delta)}{2n_2}} + \frac{1}{n_2} \right) \end{aligned}$$

By applying the basic inequality $\sqrt{a+b+c} \leq \sqrt{a} + \sqrt{b} + \sqrt{c}$ and $(x+y)^p \leq 2^{(p-1)+}(x^p + y^p)$ to decouple the terms and using Corollary G.3, we abstract the multiplicative constants (including $e^{2\kappa}$, C_0 , C_1 , and $e^{\tau_{\alpha,t}^*}$) into the asymptotic notation to obtain the final explicit finite-sample bound

$$\mathbb{P}_1(\widehat{\mathcal{R}}_t) - \mathbb{P}_1(\mathcal{R}_t^*) \leq \mathcal{O}\left(\left((1-c\rho)^t + \sqrt{\rho \log(4T/\delta)} + \eta^2/\rho^2\right)^{\frac{1+\bar{\gamma}}{2}} + \sqrt{\frac{\log(2/\delta)}{n_2}} + \frac{1}{n_2}\right).$$

When $\Delta_{n_2}(\delta) \leq 1$, the condition $\underline{\gamma} \leq 1 + \bar{\gamma}$ implies $\Delta_{n_2}(\delta)^{(1+\bar{\gamma})/\underline{\gamma}} \leq \Delta_{n_2}(\delta)$. When $\Delta_{n_2}(\delta) > 1$, the claimed bound is trivial since the excess Type-II error is at most one. This completes the proof. $\square$

G.4 Proof of Theorem G.2

Proof of Theorem G.2. Fix $c \in \{0, 1\}$ and suppress the class index throughout the proof. Write

$$\boldsymbol{\mu}_t := \boldsymbol{\mu}_{c,t}, \quad \widehat{\boldsymbol{\mu}}_t := \widehat{\boldsymbol{\mu}}_{c,t}, \quad \bar{\mathbf{z}}_t := \bar{\mathbf{z}}_{c,t}(\mathbf{v}_t), \quad A := A_d(\kappa).$$

By Assumption 1,

$$\bar{\mathbf{z}}_t = A\boldsymbol{\mu}_t + \boldsymbol{\zeta}_{t+1}, \quad \mathbb{E}[\boldsymbol{\zeta}_{t+1} \mid \mathcal{F}_t] = 0, \quad \|\boldsymbol{\zeta}_{t+1}\| \leq 2 \quad \text{a.s.}$$

Hence

$$\mathbf{z}_{t+1} = (1-\rho)\widehat{\boldsymbol{\mu}}_t + \rho A\boldsymbol{\mu}_t + \rho\boldsymbol{\zeta}_{t+1}, \quad \widehat{\boldsymbol{\mu}}_{t+1} = \frac{\mathbf{z}_{t+1}}{\|\mathbf{z}_{t+1}\|}.$$

Define

$$\mathbf{e}_t := \widehat{\boldsymbol{\mu}}_t - \boldsymbol{\mu}_t, \quad \mathbf{y}_t := (1-\rho)\widehat{\boldsymbol{\mu}}_t + \rho A\boldsymbol{\mu}_t.$$

Step 1: Bootstrap region and deterministic bounds. Define the bootstrap event

$$\mathcal{E}_t := \left\{ \|\mathbf{e}_s\| \leq \frac{1}{4} \text{ for all } 0 \leq s \leq t \right\}.$$

Clearly, $\mathcal{E}_t \in \mathcal{F}_t := \sigma(\{\boldsymbol{\zeta}_s\}_{s=1}^t)$ is $\mathcal{F}_t$ -measurable. On $\mathcal{E}_t$, Lemma G.8 yields

$$\|\Pi(\mathbf{y}_t) - \boldsymbol{\mu}_t\| \leq (1-c_0\rho)\|\mathbf{e}_t\|,$$

for some $c_0 \in (0, 1)$ and all $0 \leq t \leq T-1$, provided $\rho \leq \rho_0$ is small enough.

Also, on $\mathcal{E}_t$, $\langle \widehat{\boldsymbol{\mu}}_t, \boldsymbol{\mu}_t \rangle = 1 - \frac{1}{2}\|\mathbf{e}_t\|^2 \geq 1 - \frac{1}{32} = \frac{31}{32}$. Therefore,

$$\|\mathbf{y}_t\| \geq \langle \mathbf{y}_t, \boldsymbol{\mu}_t \rangle = (1-\rho)\langle \widehat{\boldsymbol{\mu}}_t, \boldsymbol{\mu}_t \rangle + \rho A \geq (1-\rho)\frac{31}{32} + \rho A.$$

Shrinking ρ_0 if necessary, we may ensure $\|\mathbf{y}_t\| \geq \frac{1}{2}$, $\forall 0 < \rho \leq \rho_0$. Since $\mathbf{z}_{t+1} = \mathbf{y}_t + \rho\boldsymbol{\zeta}_{t+1}$ and $\|\boldsymbol{\zeta}_{t+1}\| \leq 2$, we further obtain $\|\mathbf{z}_{t+1}\| \geq \|\mathbf{y}_t\| - \rho\|\boldsymbol{\zeta}_{t+1}\| \geq \frac{1}{2} - 2\rho \geq \frac{1}{4}$ for ρ_0 smaller if needed. In short, we show that as long as $\|\mathbf{e}_t\| \leq \frac{1}{4}$, we then have $\|\mathbf{y}_t\| \geq \frac{1}{2}$, $\|\mathbf{z}_{t+1}\| \geq \frac{1}{4}$, and $\langle \widehat{\boldsymbol{\mu}}_t, \boldsymbol{\mu}_t \rangle \geq 0$.

Step 2: One-step square-error decomposition. For any $\gamma > 0$, Young's inequality gives

$$\|\mathbf{e}_{t+1}\|^2 = \|\widehat{\boldsymbol{\mu}}_{t+1} - \boldsymbol{\mu}_{t+1}\|^2 \leq (1+\gamma)\|\widehat{\boldsymbol{\mu}}_{t+1} - \boldsymbol{\mu}_t\|^2 + \left(1 + \frac{1}{\gamma}\right)\|\boldsymbol{\mu}_{t+1} - \boldsymbol{\mu}_t\|^2. \quad (15)$$

By Assumption 2, $\|\boldsymbol{\mu}_{t+1} - \boldsymbol{\mu}_t\| \leq C_\mu\eta$. We now analyze the first term of (15):

$$\|\widehat{\boldsymbol{\mu}}_{t+1} - \boldsymbol{\mu}_t\|^2 = \|\Pi(\mathbf{z}_{t+1}) - \boldsymbol{\mu}_t\|^2.$$

For simplicity, we set $\mathbf{a}_t := \Pi(\mathbf{y}_t) - \boldsymbol{\mu}_t$. Then

$$\Pi(\mathbf{z}_{t+1}) - \boldsymbol{\mu}_t = \mathbf{a}_t + (\Pi(\mathbf{y}_t + \rho\boldsymbol{\zeta}_{t+1}) - \Pi(\mathbf{y}_t)).$$

Hence, it follows that

$$\begin{aligned} \|\Pi(\mathbf{z}_{t+1}) - \boldsymbol{\mu}_t\|^2 &= \|\mathbf{a}_t\|^2 + 2\langle \mathbf{a}_t, \Pi(\mathbf{y}_t + \rho\boldsymbol{\zeta}_{t+1}) - \Pi(\mathbf{y}_t) \rangle \\ &\quad + \|\Pi(\mathbf{y}_t + \rho\boldsymbol{\zeta}_{t+1}) - \Pi(\mathbf{y}_t)\|^2. \end{aligned} \quad (16)$$